A Few Notes on Vector Calculus

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1 Introduction

TBD

2 Notation

Some of the notation we'll use is described in this section.

2.1 Vectors

Some authors use bold font to denote a vector. Adding a "hat" denotes a unit vector. A unit vector $\hat{\mathbf{u}}$ is defined to be

$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$$

that is, $\hat{\mathbf{u}}$ is a vector of unit length in the $\mathbf{u}$ direction.

For example $\hat{\mathbf{i}}$ denotes the unit vector in the x direction. Using these definitions the parametric form of some C in $\mathbb{R}^3$, $\mathbf{r}(t)$, would be

$$\mathbf{r}(t) = g(t)\hat{\mathbf{i}} + h(t)\hat{\mathbf{j}} + k(t)\hat{\mathbf{k}}$$

where $g(t)$, $h(t)$ and $k(t)$ are scalar functions and $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$, the basis vectors in $\mathbb{R}^3$, have column vector format

$$\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},\ \hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix},\ \hat{\mathbf{k}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Another common notation is

$$\vec{r}(t) = g(t)\hat{i} + h(t)\hat{j} + k(t)\hat{k}$$

In this document I'll use these notations interchangeably.

3 Definitions and a Few Theorems

Definition 3.1. Dot Product: The dot product (aka scalar product or inner product) of two column vectors $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$, where $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, is defined to be the scalar value

$$\mathbf{a} \cdot \mathbf{b} = \langle \mathbf{a}, \mathbf{b} \rangle = \sum_{j=1}^{n} a_j b_j = \mathbf{a}^\intercal \mathbf{b}$$

According to the Pythagorean theorem, the length of a vector $\mathbf{a} = (a_1, a_2, a_3) \in \mathbb{R}^3$, denoted $\|\mathbf{a}\|$, equals $\sqrt{a_1^2 + a_2^2 + a_3^2}$. The next definition is a generalization of the notion of length to vectors in $\mathbb{R}^n$.

Definition 3.2. Euclidean Norm: The Euclidean norm of a vector

$$\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \mathbb{R}^n$$

is defined to be

$$\|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle} = \sqrt{\sum_{j=1}^{n} a_j^2} \tag{1}$$

Theorem 3.1. Cauchy–Schwarz Inequality:

$$\big|\langle \mathbf{a}, \mathbf{b} \rangle\big| \leq \|\mathbf{a}\| \|\mathbf{b}\|$$

Proof: The inequality is trivial if either $\mathbf{a}$ or $\mathbf{b}$ is zero, so assume that neither is. Now, let $\mathbf{x} = \frac{\mathbf{a}}{\|\mathbf{a}\|}$ and $\mathbf{y} = \frac{\mathbf{b}}{\|\mathbf{b}\|}$ and so $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$ ($\mathbf{x} = \hat{\mathbf{a}}$ and $\mathbf{y} = \hat{\mathbf{b}}$).

Then one way to prove the Cauchy–Schwarz inequality is to use the reverse triangle inequality:

$$\big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| \leq \|\mathbf{x} - \mathbf{y}\| \tag{2}$$

We can get the reverse triangle inequality from the triangle inequality [4, 7] by observing that

$$\begin{aligned}
\|\mathbf{x}\| &= \|\mathbf{x} + (-\mathbf{y} + \mathbf{y})\| && \text{\# add 0 to } \mathbf{x}\text{: } \mathbf{x} = \mathbf{x} + 0 = \mathbf{x} + (-\mathbf{y} + \mathbf{y}) \\
&= \|(\mathbf{x} - \mathbf{y}) + \mathbf{y}\| && \text{\# addition is associative: } \mathbf{v}_1 + (\mathbf{v}_2 + \mathbf{v}_3) = (\mathbf{v}_1 + \mathbf{v}_2) + \mathbf{v}_3 \\
&\leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| && \text{\# by the triangle inequality: } \|\mathbf{v}_1 + \mathbf{v}_2\| \leq \|\mathbf{v}_1\| + \|\mathbf{v}_2\| \\
&\Rightarrow \|\mathbf{x}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| && \text{\# combine expressions for } \|\mathbf{x}\| \\
&\Rightarrow \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| && \text{\# subtract } \|\mathbf{y}\| \text{ from both sides}
\end{aligned}$$

and

$$
\begin{aligned}
\|\mathbf{y}\| &= \|\mathbf{y} + (-\mathbf{x} + \mathbf{x})\| && \text{\# add 0 to } \mathbf{y}\text{: } \mathbf{y} = \mathbf{y} + 0 = \mathbf{y} + (-\mathbf{x} + \mathbf{x}) \\
&= \|(\mathbf{y} - \mathbf{x}) + \mathbf{x}\| && \text{\# addition is associative} \\
&\leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| && \text{\# by the triangle inequality} \\
&\Rightarrow \|\mathbf{y}\| \leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| && \text{\# combine expressions for } \|\mathbf{y}\| \\
&\Rightarrow \|\mathbf{y}\| - \|\mathbf{x}\| \leq \|\mathbf{y} - \mathbf{x}\| && \text{\# subtract } \|\mathbf{x}\| \text{ from both sides} \\
&\Rightarrow -\Big[\|\mathbf{y}\| - \|\mathbf{x}\|\Big] \geq -\|\mathbf{y} - \mathbf{x}\| && \text{\# multiply both sides by -1: } -1 * (\mathbf{v}_1 \leq \mathbf{v}_2) \Rightarrow -\mathbf{v}_1 \geq -\mathbf{v}_2 \\
&\Rightarrow \|\mathbf{x}\| - \|\mathbf{y}\| \geq -\|\mathbf{x} - \mathbf{y}\| && \text{\# } \|\mathbf{y} - \mathbf{x}\| = \|-(\mathbf{y} - \mathbf{x})\| = \|\mathbf{x} - \mathbf{y}\| \text{ by Equation (1)} \\
&\Rightarrow -\|\mathbf{x} - \mathbf{y}\| \leq \|\mathbf{x}\| - \|\mathbf{y}\| && \text{\# rearrange}
\end{aligned}
$$

Combining these two statements gives us the reverse triangle inequality (Equation (2)):

$$
-\|\mathbf{x} - \mathbf{y}\| \leq \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| \Rightarrow \big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| \leq \|\mathbf{x} - \mathbf{y}\|
$$

Using the reverse triangle inequality we can see that

$$
\begin{aligned}
\big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| &\leq \|\mathbf{x} - \mathbf{y}\| && \text{\# reverse triangle inequality (Equation (2))} \\
&\Rightarrow 0 \leq \|\mathbf{x} - \mathbf{y}\| && \text{\# } \|\mathbf{x}\| = \|\mathbf{y}\| = 1 \text{ by definition above, } 1 - 1 = 0,\ |0| = 0 \\
&\Rightarrow 0 \leq \|\mathbf{x} - \mathbf{y}\|^2 && \text{\# } \|\mathbf{x} - \mathbf{y}\| \geq 0 \Rightarrow \|\mathbf{x} - \mathbf{y}\|^2 \geq 0 \\
&\Rightarrow 0 \leq \sum_{j=1}^{n} (x_j - y_j)^2 && \text{\# } \|\mathbf{x} - \mathbf{y}\|^2 = \sum_{j=1}^{n} (x_j - y_j)^2 \text{ by Equation (1)} \\
&\Rightarrow 0 \leq \sum_{j=1}^{n} \left(x_j^2 - 2x_j y_j + y_j^2\right) && \text{\# expand previous line} \\
&\Rightarrow 0 \leq \sum_{j=1}^{n} x_j^2 - 2\sum_{j=1}^{n} x_j y_j + \sum_{j=1}^{n} y_j^2 && \text{\# sum is a linear operator} \\
&\Rightarrow 0 \leq \|\mathbf{x}\|^2 - 2\langle \mathbf{x}, \mathbf{y}\rangle + \|\mathbf{y}\|^2 && \text{\# } \|\mathbf{v}\|^2 = \sum_{j=1}^{n} v_j^2 \text{ and } \langle \mathbf{x}, \mathbf{y}\rangle = \sum_{j=1}^{n} x_j y_j \text{ by Equation (1)} \\
&\Rightarrow 0 \leq 1 - 2\langle \mathbf{x}, \mathbf{y}\rangle + 1 && \text{\# } \|\mathbf{x}\| = \|\mathbf{y}\| = 1 \Rightarrow \|\mathbf{x}\|^2 = \|\mathbf{y}\|^2 = 1 \\
&\Rightarrow 0 \leq 2 - 2\langle \mathbf{x}, \mathbf{y}\rangle && \text{\# } 1 + 1 = 2 \\
&\Rightarrow -2 \leq -2\langle \mathbf{x}, \mathbf{y}\rangle && \text{\# subtract 2 from both sides} \\
&\Rightarrow 1 \geq \langle \mathbf{x}, \mathbf{y}\rangle && \text{\# divide by -2, reversing inequality} \\
&\Rightarrow \langle \mathbf{x}, \mathbf{y}\rangle \leq 1 && \text{\# rearrange} \\
&\Rightarrow \left\langle \frac{\mathbf{a}}{\|\mathbf{a}\|}, \frac{\mathbf{b}}{\|\mathbf{b}\|} \right\rangle \leq 1 && \text{\# } \mathbf{x} = \frac{\mathbf{a}}{\|\mathbf{a}\|},\ \mathbf{y} = \frac{\mathbf{b}}{\|\mathbf{b}\|} \text{ by definition above} \\
&\Rightarrow \frac{\langle \mathbf{a}, \mathbf{b}\rangle}{\|\mathbf{a}\|\|\mathbf{b}\|} \leq 1 && \text{\# scalar multiplication rule: } \langle c_1\mathbf{v}_1, c_2\mathbf{v}_2\rangle = c_1 c_2 \langle \mathbf{v}_1, \mathbf{v}_2\rangle \\
&\Rightarrow \langle \mathbf{a}, \mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\| && \text{\# multiply both sides by } \|\mathbf{a}\|\|\mathbf{b}\|
\end{aligned}
$$

If we replace $\mathbf{a}$ by $-\mathbf{a}$ we see by Equation (1) that $\|-\mathbf{a}\| = \|\mathbf{a}\|$. Further, by the scalar multiplication rule for dot products with $c_1 = -1$ and $c_2 = 1$ we have $\langle -\mathbf{a}, \mathbf{b}\rangle = -\langle \mathbf{a}, \mathbf{b}\rangle$. Taken together these results imply that[1]

$$-\langle \mathbf{a}, \mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$$

Next note that the absolute value of x, $|x|$, is defined to be

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0. \end{cases} \tag{3}$$

So one way to see the Cauchy–Schwarz inequality is by setting x equal to $\langle \mathbf{a}, \mathbf{b}\rangle$ in Equation (3). Then when $\langle \mathbf{a}, \mathbf{b}\rangle \geq 0$ we have $\left|\langle \mathbf{a}, \mathbf{b}\rangle\right| = \langle \mathbf{a}, \mathbf{b}\rangle$ and we showed above that $\langle \mathbf{a}, \mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$. Similarily, when $\langle \mathbf{a}, \mathbf{b}\rangle < 0$ we have $\left|\langle \mathbf{a}, \mathbf{b}\rangle\right| = -\langle \mathbf{a}, \mathbf{b}\rangle$ and we showed that $-\langle \mathbf{a}, \mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$. Together these give us $\left|\langle \mathbf{a}, \mathbf{b}\rangle\right| \leq \|\mathbf{a}\|\|\mathbf{b}\|$, namely, the Cauchy–Schwarz inequality. $\square$

3.1 Derivatives

We will also denote the derivative of a vector $\mathbf{r}$ with respect to t by $\dot{\mathbf{r}}$, $\dfrac{d\mathbf{r}}{dt}$ or $\mathbf{r}'(t)$. Note also that by definition

$$\frac{d\mathbf{r}}{dt} = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

3.2 Vector Fields

In vector calculus and physics, a vector field is an assignment of a vector to each point in a subset of space [2]. For example, a vector field in the plane can be visualized as a collection of arrows with a given magnitude and direction, each attached to a point in the plane. Vector fields are often used to model the speed and direction of a moving fluid throughout space, or the strength and direction of some force, such as the magnetic or gravitational force, as it changes from one point to another point.

For example, the wind velocity vector field for the 2011 Joplin, MO tornado [5] is shown in Figure 1. Here the color represents the wind speed $\|\mathbf{v}\|$, where $\mathbf{v}$ is the wind velocity vector.

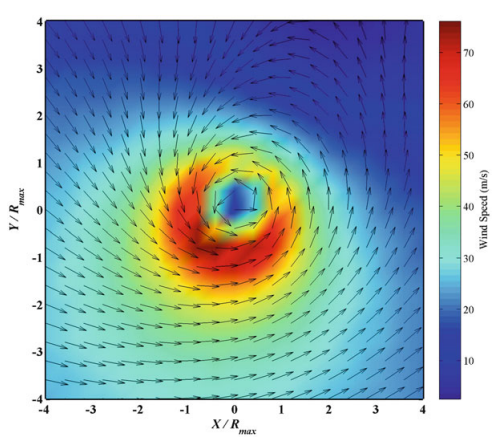

Figure 1: 2011 Joplin, MO Tornado Wind Velocity Vector Field

[1] There is nothing special about using $\mathbf{a}$ here. We could use $\mathbf{b}$ and get the same result.

The general form of a vector field (here in three dimensions) is

$$\vec{\mathbf{F}}(x, y, z) = P(x, y, z)\,\hat{\mathbf{i}} + Q(x, y, z)\,\hat{\mathbf{j}} + R(x, y, z)\,\hat{\mathbf{k}}$$

where P, Q, and R are scalar functions. That is, they are functions f such that $f : \mathbb{R}^3 \to \mathbb{R}$.

4 Arc Length

Consider a segment of a parametric curve $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}} + k(t)\,\hat{\mathbf{k}}$ between two points $P = \mathbf{r}(t)$ and $Q = \mathbf{r}(t + \Delta t)$, as shown in Figure 2.

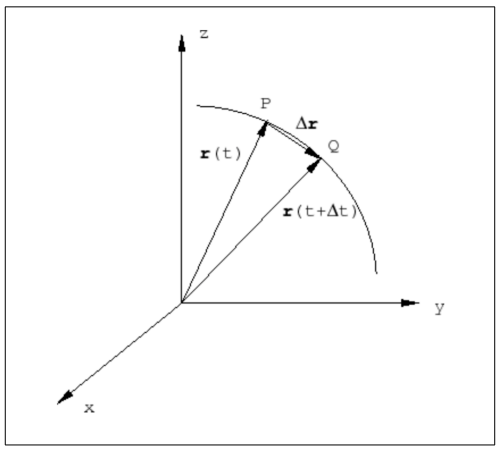

Figure 2: $\Delta\mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$

What is the length of the segment of the curve between P and Q? This length is called the arc length and is denoted by Δs. Δs can be approximated by the chord length $\|\Delta\mathbf{r}\|$. Then we can compute the infinitesimal arc length ds as follows:

$$
\begin{aligned}
\Delta s &\approx \|\Delta \mathbf{r}\| && \text{\# approximate } \Delta s \text{ by } \|\Delta \mathbf{r}\| \\
&= \|\mathbf{r}(t+\Delta t) - \mathbf{r}(t)\| && \text{\# } \Delta \mathbf{r} = \mathbf{r}(t+\Delta t) - \mathbf{r}(t) \\
&= \|\mathbf{r}(t+\Delta t) - \mathbf{r}(t)\| \cdot \frac{\Delta t}{\Delta t} && \text{\# multiply by } 1 = \frac{\Delta t}{\Delta t} \\
&= \left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# assume } \Delta t > 0 \\
\Rightarrow\ & \Delta s = \left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# combine expressions for } \Delta s\ (\approx \to = \ ?) \\
\Rightarrow\ & \lim_{\Delta t \to 0} \Delta s = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# take the limit of both sides} \\
\Rightarrow\ & ds = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# } \Delta t \to 0 \Rightarrow \Delta s \to 0 \Rightarrow \lim_{\Delta t \to 0} \Delta s = ds \\
\Rightarrow\ & ds = \lim_{\Delta t \to 0} \left\| \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# product rule for limits [1]} \\
\Rightarrow\ & ds = \left\| \lim_{\Delta t \to 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# for a normed vector space } X\ \lim \|x_n\| = \|\lim x_n\| \\
\Rightarrow\ & ds = \left\| \frac{d\mathbf{r}}{dt} \right\| \cdot dt && \text{\# } \lim_{\Delta t \to 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{d\mathbf{r}}{dt},\ \lim_{\Delta t \to 0} \Delta t = dt
\end{aligned}
$$

Interestingly we can also see this using the Pythagorean theorem. Specifically, if we move an infinitesimal distance in the x and y directions we get a triangle in the x-y plane who's sides have length dx and dy and who's hypotenuse is ds. This scenario is shown for some curve $f(x)$ in Figure 3.

In the two dimensional case where the parameterization is $x = g(t)$ and $y = h(t)$ we have $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}}$. Further, since derivative is a linear operator [6] we know that

$$
\frac{d\mathbf{r}}{dt} = \frac{dg}{dt}\,\hat{\mathbf{i}} + \frac{dh}{dt}\,\hat{\mathbf{j}} \tag{4}
$$

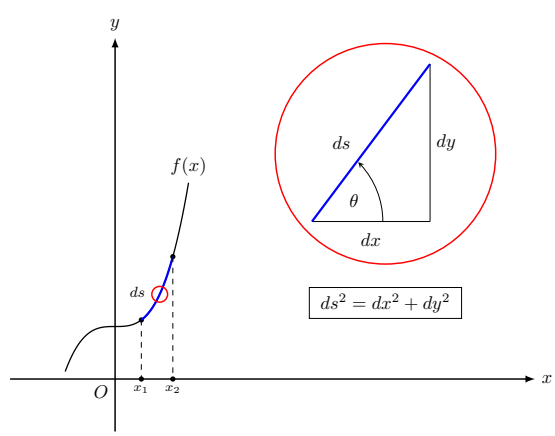

Figure 3: $f(x)$, ds and the Pythagorean Theorem

So in the source domain the Pythagorean theorem tells us that $ds = \sqrt{dx^2 + dy^2}$. Using this expression for ds we can see that

$$
\begin{aligned}
ds &= \sqrt{dx^2 + dy^2} && \text{\# Figure 3} \\
&= \sqrt{dg^2 + dh^2} && \text{\# switch to parametric form: } x = g(t) \text{ and } y = h(t) \\
&= \sqrt{dg^2 + dh^2} \cdot \frac{dt}{dt} && \text{\# multiply by } 1 = \frac{dt}{dt},\ dt > 0 \\
&= \sqrt{dg^2 + dh^2} \cdot \left[\sqrt{\frac{1}{dt^2}} \cdot dt\right] && \text{\#} \ \frac{x}{x} = \sqrt{\left(\frac{1}{x}\right)^2} \cdot x = \sqrt{\frac{1}{x^2}} \cdot x,\ x > 0 \\
&= \sqrt{\left(dg^2 + dh^2\right) \cdot \left(\frac{1}{dt^2}\right)} \cdot dt && \text{\# simplify} \\
&= \sqrt{\left(\frac{dg}{dt}\right)^2 + \left(\frac{dh}{dt}\right)^2} \cdot dt && \text{\# simplify} \\
&= \sqrt{g'(t)^2 + h'(t)^2} \cdot dt && \text{\# switch notation from Leibniz} \rightarrow \text{Lagrange: } \frac{df}{dt} = f'(t) \\
&= \left\|\frac{d\mathbf{r}}{dt}\right\| \cdot dt && \text{\# Equation (4) and the definition of } \|\cdot\|
\end{aligned}
$$

In any event, now we see that

$$
\frac{ds}{dt} = \left\|\frac{d\mathbf{r}}{dt}\right\| \tag{5}
$$

and so

$$
\frac{ds}{dt} = \|\dot{\mathbf{r}}\| \tag{6}
$$

Since $\frac{ds}{dt} = \left\|\frac{d\mathbf{r}}{dt}\right\| = \|\dot{\mathbf{r}}\|$ we also know that the unit tangent vector $\mathbf{T} = \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} = \frac{d\mathbf{r}}{ds}$ [3]. One way to see this is to notice that

$$
\begin{aligned}
\mathbf{T} &= \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} && \text{\# definition of the unit tangent vector } \mathbf{T} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\|\dot{\mathbf{r}}\|} && \text{\# definition of } \dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\left(\frac{ds}{dt}\right)} && \text{\# Equation (6): } \|\dot{\mathbf{r}}\| = \frac{ds}{dt} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\left(\frac{ds}{dt}\right)} \cdot \frac{dt}{dt} && \text{\# multiply by } 1 = \frac{dt}{dt} \\
&= \frac{d\mathbf{r}}{ds} && \text{\# simplify}
\end{aligned}
$$

5 Conclusions

Acknowledgements

LaTeX Source

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